

Parameterized Algorithms and Parameter Selection for Fast GPU–GPU Collective Communication

Peizhi Liu Sean Rhee Michael Wilkins Peter Dinda

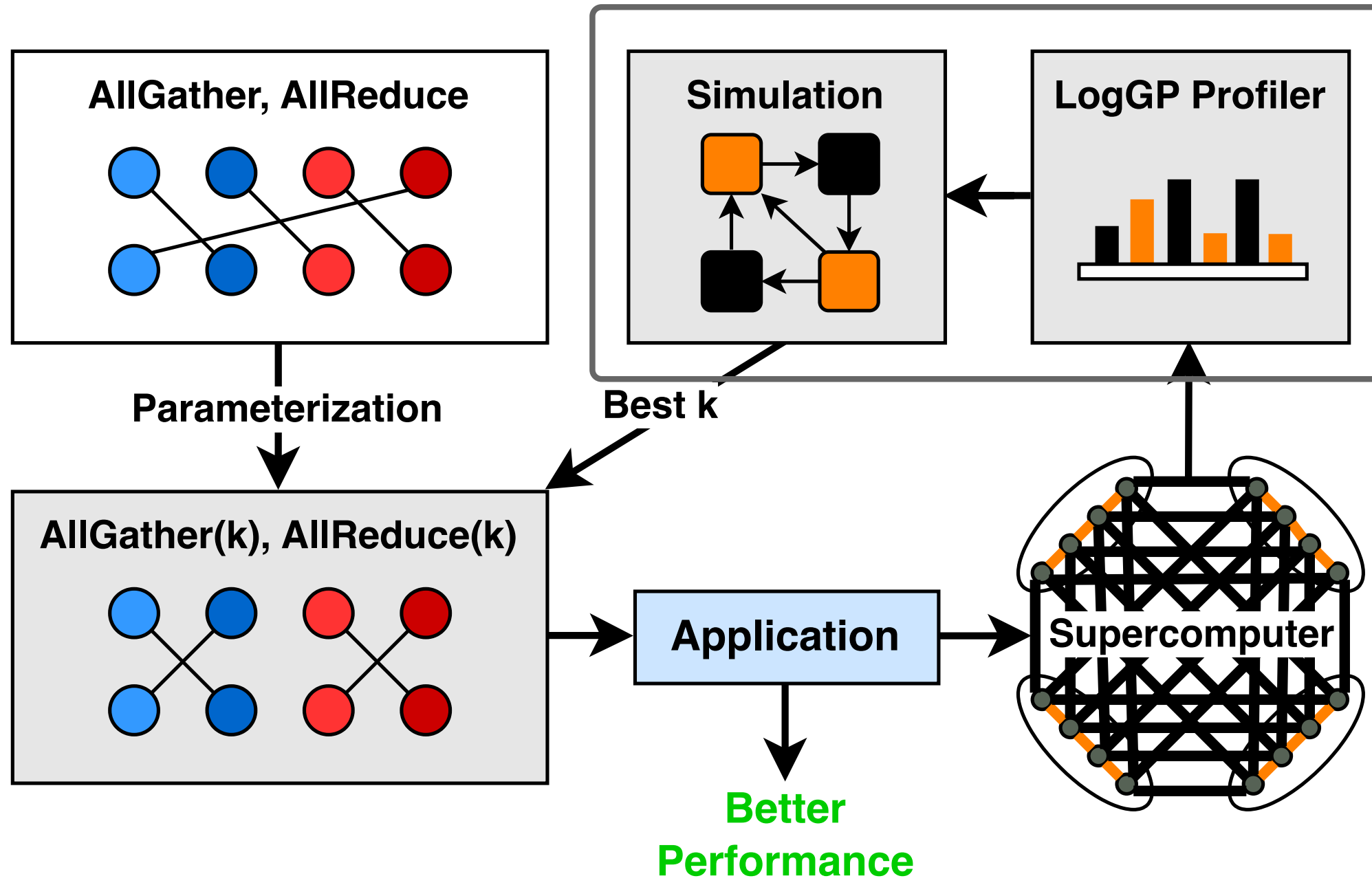
Inefficient Communication on Supercomputers

Collective communication (collectives) is a major bottleneck in many AI and scientific applications running on GPU supercomputers. Current algorithms that implement collectives like **AllGather** and **AllReduce** are not suited for today's supercomputers for the following reasons:

- Today's supercomputers feature **complex interconnect topologies**.
- Optimized bespoke algorithms are **not portable** across machines.

Is it possible to achieve **performant** collective communication without sacrificing **portability** on modern supercomputers?

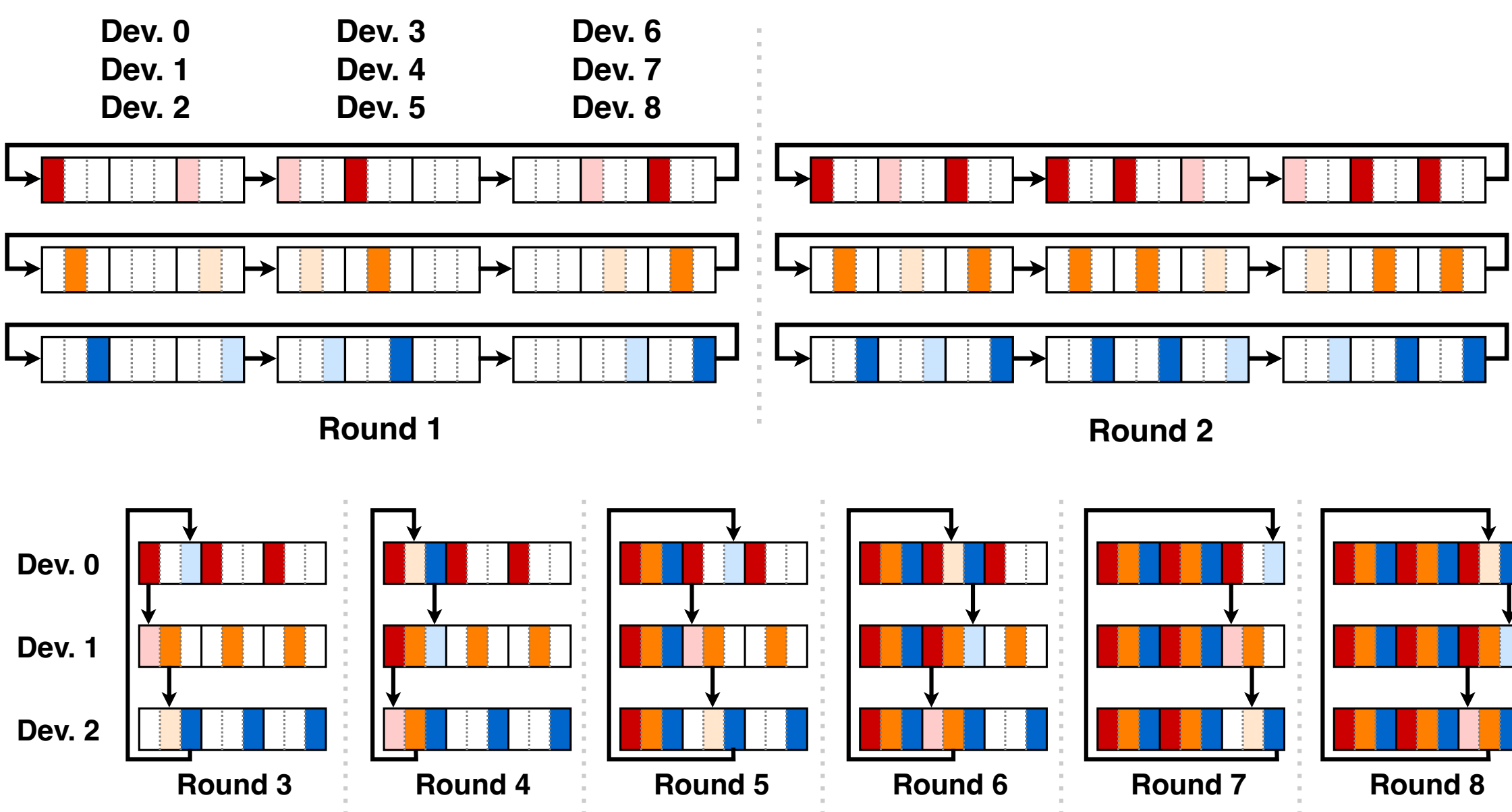
Our Solution: Parameterized Algorithms + Simulation



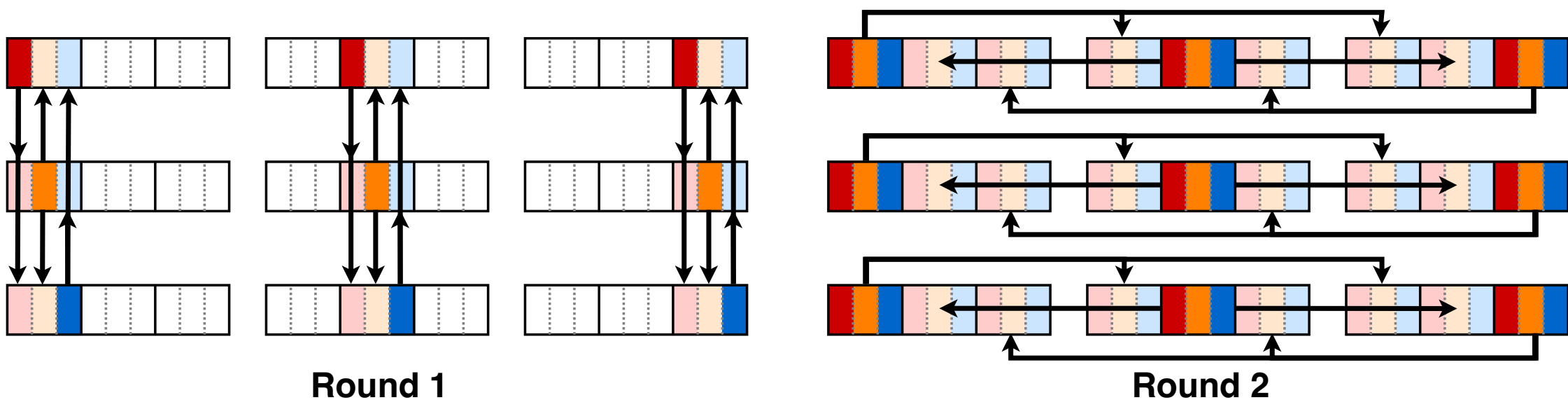
Parameterized Algorithms for GPU Collectives

Parameterization takes an existing collective algorithm and adds a **degree of freedom, k** , to it. What k means depends on the algorithm.

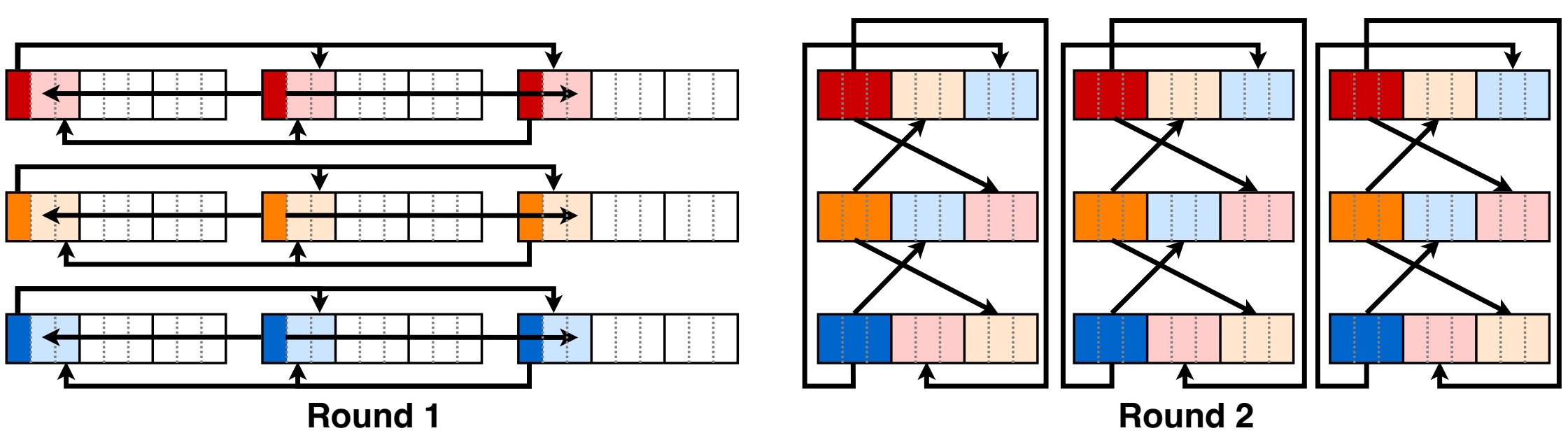
K-Ring parameterizes the ring algorithm. k is the number of devices in a sub-ring. Algorithm is composed of intra- and inter-sub-ring communication rounds. More data is sent in the sub-ring between local devices.



Recursive Multiplying generalizes recursive doubling. k_1 is the number of partners a device exchanges data with in a round. For each subsequent round, the offset between partners increases by a factor of k .



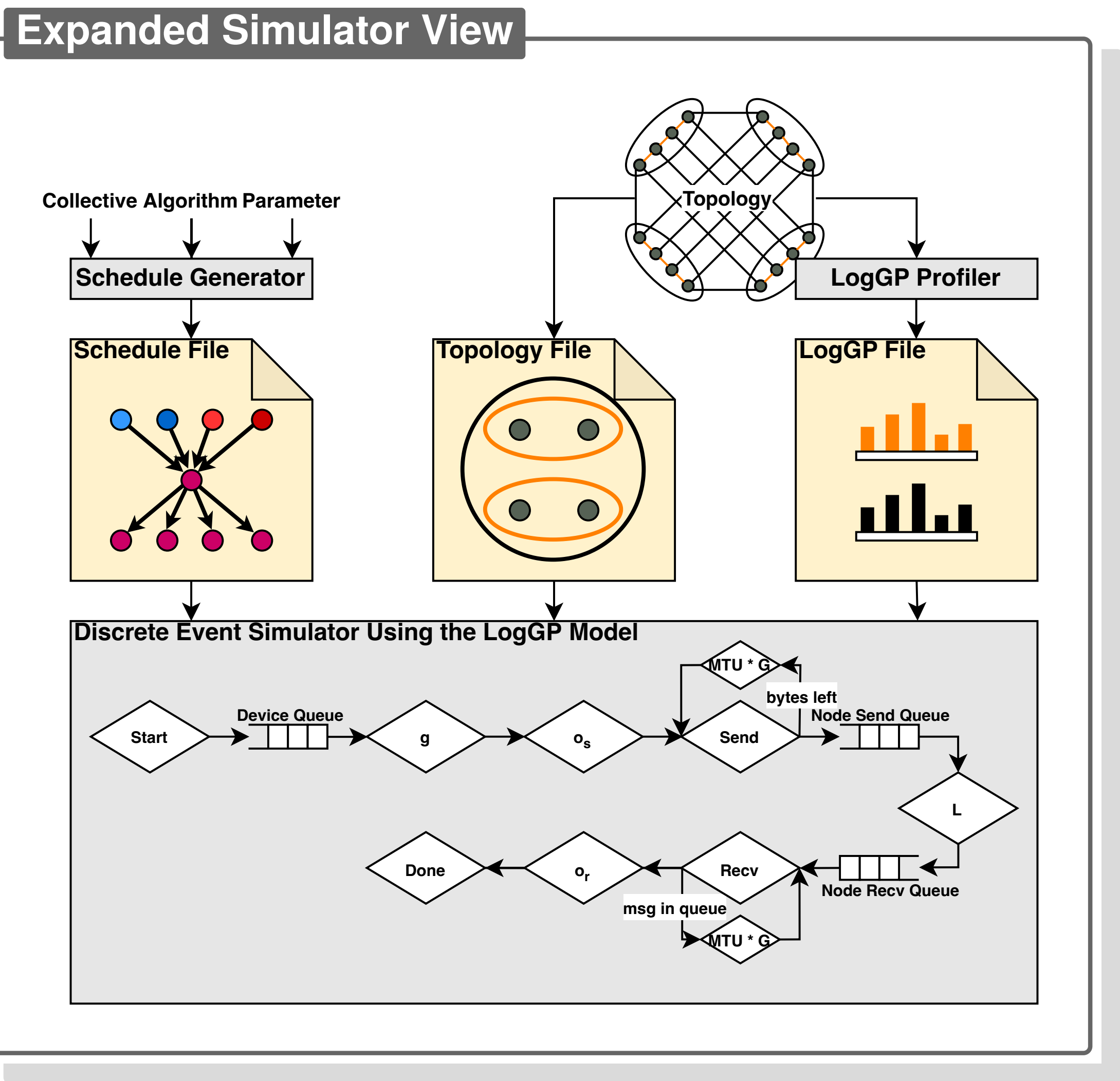
Permuted Recursive Multiplying optimizes recursive multiplying by sending smaller messages over longer distances first, then restores buffer order with a **GPU-local permutation kernel**.



The above three parameterized algorithms were implemented for both **AllGather** and **AllReduce**.

Fast Parameter Selection Through Simulation

Discrete event simulation is used to quickly select the best parameter for each collective offline without exhaustive measurements. **Node and device-level contention modeling** is crucial for accuracy.

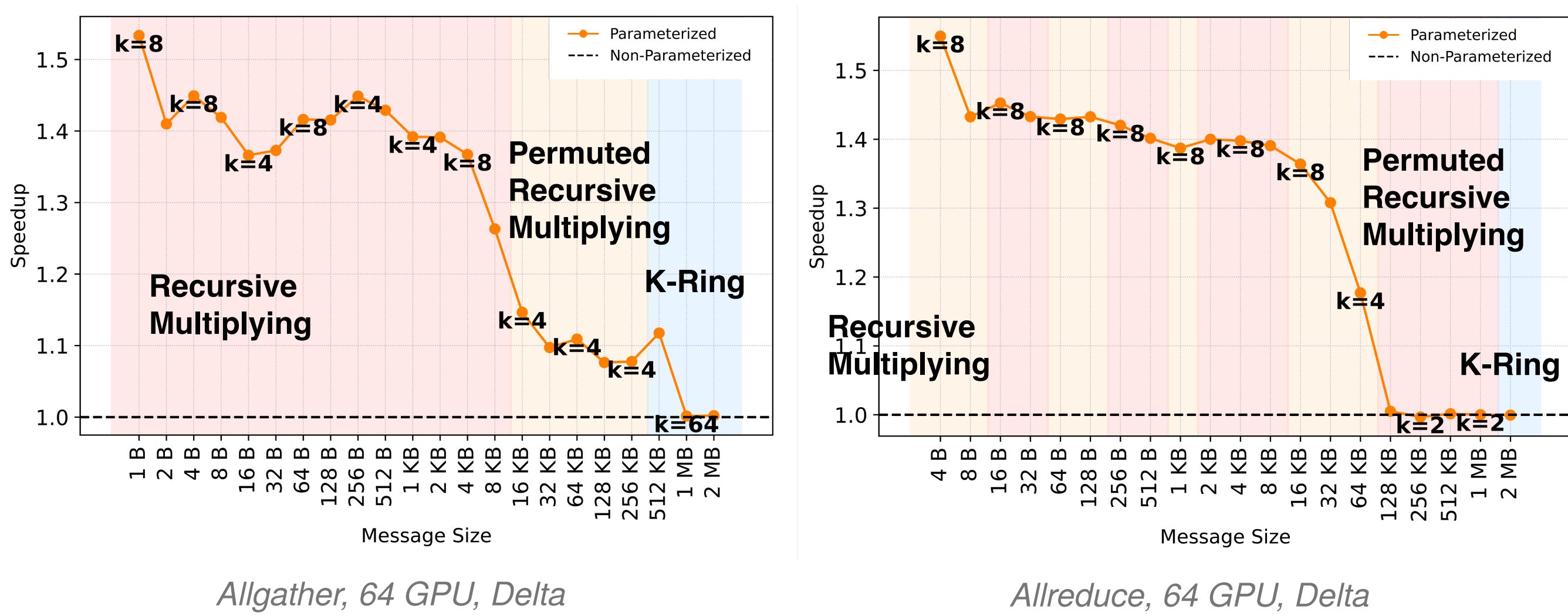


Implementation and Evaluation

- LD_PRELOAD** library overriding NCCL *Allgather* and *Allreduce*.
- Built on top of NCCL send/receive primitives.
- Evaluated on NCSA **Delta** and Argonne **Polaris**: 4 NVIDIA A100 GPUs per node with NVLink and HPE/Cray Slingshot 11.

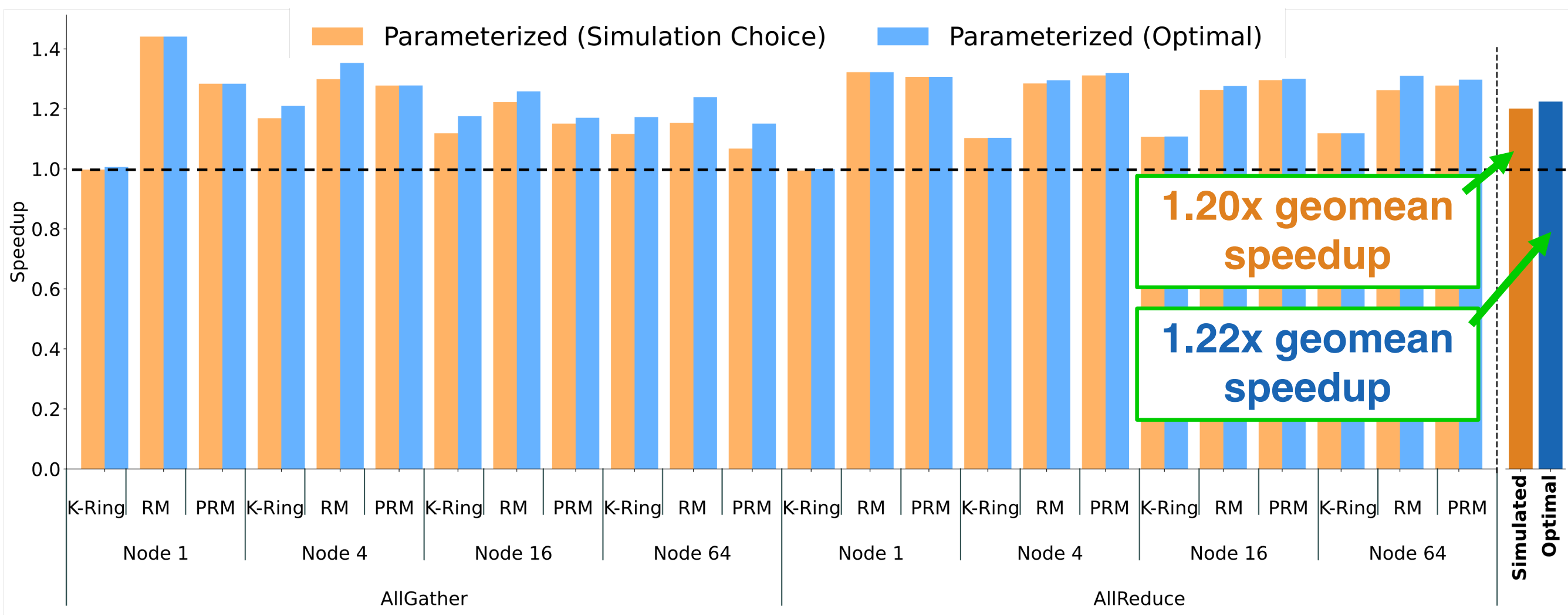
Measured Speedups on Real GPU Supercomputers

Parameterized algorithms improve performance across message sizes, achieving up to **50% speedup** over non-parameterized algorithms for both **AllGather** and **AllReduce** on Delta with 64 GPUs.



Simulation Makes Parameterization Practical

Simulation-derived parameters achieve **98% of optimal performance** on Delta across various scales, collectives, algorithms, and message sizes.



Takeaway

Parameterized collective algorithms with **simulation-based parameter selection** provides a **performant** and **portable** path towards efficient communication on GPU supercomputers.



Paper